

**CAFFEINE & CADAVER**

MBBS BIOCHEMISTRY NOTES SERIES

PYQ 2010–2025

MBBS BIOCHEMISTRY · MODULE 03

Lipid Metabolism

β-Oxidation · Ketone Bodies · Cholesterol · Lipoproteins · Fatty Acid Synthesis

SUBJECT**Biochemistry — Paper 1****EXAM RELEVANCE****Group A · B · C · D****PYQ FREQUENCY****10+ times (2010–2025)**

★ MBBS PYQ APPEARANCES – YEARS

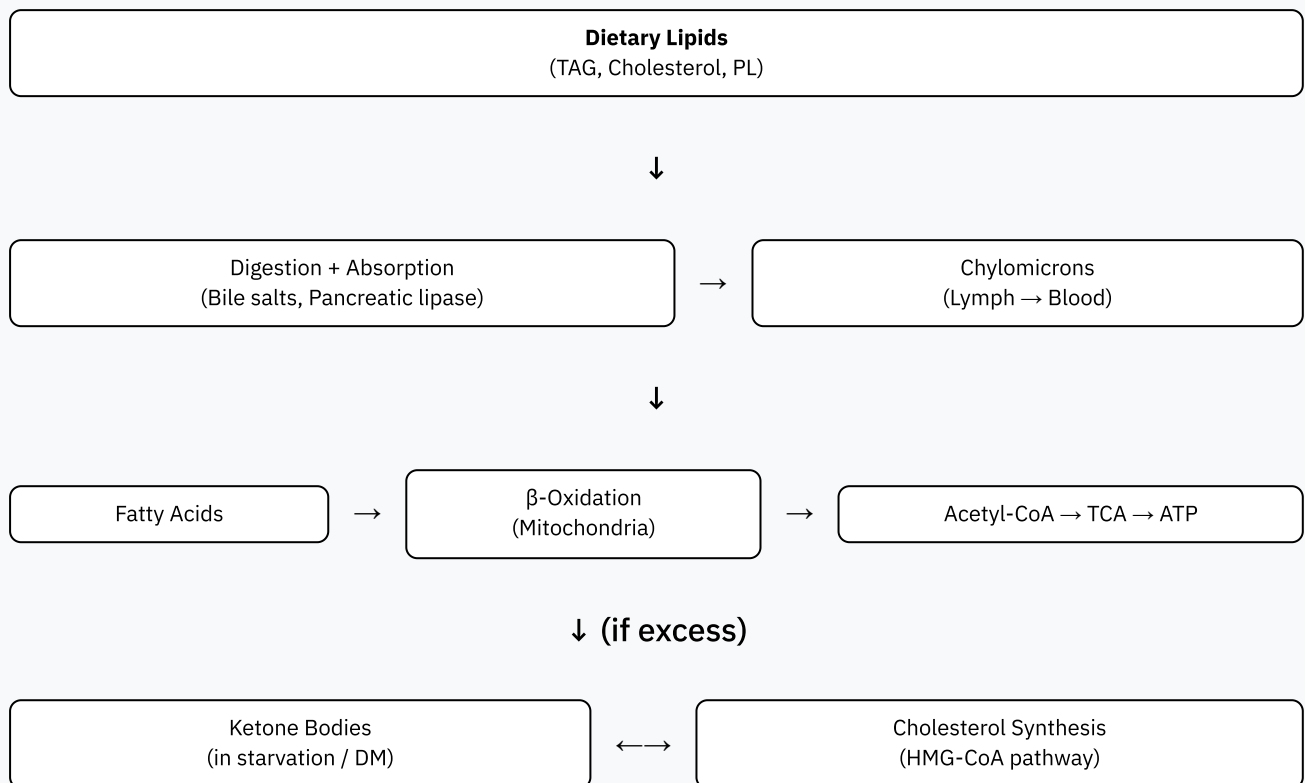
2010 2010-S 2011-S 2012-S 2013 2013-S 2014 2015 2015-S 2017 2017-S 2018
2019 2019-S 2021 2022 2023 2024 2025

01 The Big Picture

🔗 WHY THIS MATTERS CLINICALLY

Lipid metabolism is where fat becomes energy, where drugs like **statins** intervene, and where diseases like **atherosclerosis, fatty liver, DKA, and familial hypercholesterolaemia** originate. The examiner loves lipid questions because they demand pathway knowledge, enzyme names, and clinical application all at once — often in a case-based Group A.

// MASTER OVERVIEW – FATE OF LIPIDS



02 β -Oxidation of Fatty Acids ★ EXAMINER FAV

★ ONE-LINER

Sequential removal of **2-carbon units** as Acetyl-CoA from the β -carbon of a fatty acid, occurring in the **mitochondrial matrix**, generating $\text{FADH}_2 + \text{NADH}$ per cycle.

// CARNITINE SHUTTLE – ENTRY INTO MITOCHONDRIA

Fatty Acid (cytoplasm)

→ **Acyl-CoA Synthetase (ATP)**

Fatty Acyl-CoA

↓ **CPT-I** RLS (outer membrane)

Acyl-Carnitine

→ **Translocase (inner membrane)**

Acyl-Carnitine (matrix)

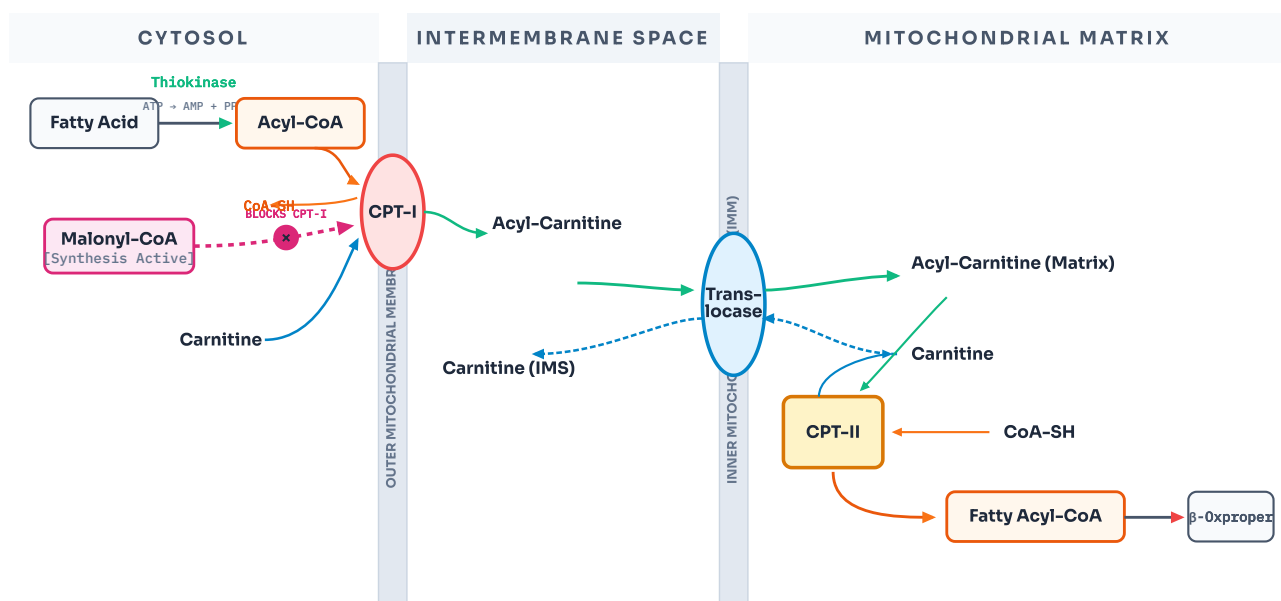
↓ **CPT-II (inner membrane)**

Fatty Acyl-CoA (matrix)

→

β -Oxidation ✓

FIG 3.1: TEXTBOOK-GRADE CARNITINE SHUTTLE & REGULATION BY MALONYL-COA





CPT-I is the rate-limiting step RLS of β -oxidation. Inhibited by **Malonyl-CoA** (the first committed intermediate of fatty acid synthesis) — this is how the cell prevents simultaneous synthesis and breakdown of fatty acids.

// ONE ROUND OF β -OXIDATION (PALMITATE C16 EXAMPLE)

Palmitoyl-CoA (C16)

↓ Step 1: Acyl-CoA Dehydrogenase → FADH_2

Trans- Δ^2 -Enoyl-CoA

↓ Step 2: Enoyl-CoA Hydratase → H_2O added

L-3-Hydroxyacyl-CoA

↓ Step 3: Hydroxyacyl-CoA Dehydrogenase → NADH

3-Ketoacyl-CoA

↓ Step 4: Thiolase → CoA added

Acetyl-CoA (C2) ✓

+

Acyl-CoA (C14) → next round

Palmitate (C16):

7 rounds of β -oxidation

8 Acetyl-CoA produced

Per round:

1 FADH_2 + 1 NADH

= 4 ATP

Net ATP (C16):

106 ATP total

(−2 for activation)

■ LYNEN'S SPIRAL (4-STEP CYCLE) & MOLECULAR ENERGETICS

β -Oxidation of fatty acids is also called **Lynen's spiral**. Long-chain fatty acids are sequentially degraded in a repeating 4-step cyclical spiral process to release 2-carbon Acetyl-CoA units.

Step 1: Dehydrogenation (FAD-dependent)

Acyl-CoA is oxidized by mitochondrial membrane-bound **Acyl-CoA Dehydrogenase** to form trans- Δ^2 -Enoyl-CoA. Electrons are transferred to FAD $\rightarrow$ FADH₂ which feeds into the electron transport chain (yielding **1.5 ATP**).

Step 2: Hydration

Hydration of the double bond by **Enoyl-CoA Hydratase** adds water across the trans double bond, yielding L-3-Hydroxyacyl-CoA.

Step 3: Dehydrogenation (NAD-dependent)

L-3-Hydroxyacyl-CoA is oxidized by **3-Hydroxyacyl-CoA Dehydrogenase** to yield 3-Ketoacyl-CoA, generating NADH + H⁺ (yielding **2.5 ATP** in ETC).

Step 4: Thiolitic Cleavage (Thiolase)

Cleavage of 3-Ketoacyl-CoA by **Thiolase (needs CoASH)** releases 1 molecule of **Acetyl-CoA** and a Fatty Acyl-CoA molecule that is **2 carbons shorter**. The shortened Fatty Acyl-CoA re-enters the cycle.

12
34

Pathway Energetics Calculations:

- **Acetyl-CoA Yield** = $n / 2$ (each enters TCA cycle, yielding 10 ATP).
- **Number of Cycles** = $(n / 2) - 1$ (each cycle generates 1 FADH₂ + 1 NADH = 4 ATP).
- **Activation Cost** = 2 ATP equivalents (Thiokinase reaction consumes 1 ATP $\rightarrow$ AMP + PPi, followed by immediate pyrophosphatase hydrolysis).
- **Palmitate (16C)**: 8 Acetyl-CoA (80 ATP) + 7 cycles (28 ATP) = 108 ATP. Net = **106 ATP**.
- **Stearate (18C)**: 9 Acetyl-CoA (90 ATP) + 8 cycles (32 ATP) = 122 ATP. Net = **120 ATP**.

💡 SPECIALTY FATTY ACID OXIDATIONS

1. Odd-Chain Fatty Acid Oxidation:

Odd-chain fatty acids undergo normal β -oxidation until the final thiolitic cleavage yields one molecule of 3-carbon **Propionyl-CoA** instead of two Acetyl-CoAs. Propionyl-CoA $\rightarrow$ **D-Methylmalonyl-CoA** (via Biotin-dependent *Propionyl-CoA Carboxylase*, consuming ATP and CO_2) $\rightarrow$ L-Methylmalonyl-CoA $\rightarrow$ **Succinyl-CoA** (via Vitamin B12-dependent *Methylmalonyl-CoA Mutase*) $\rightarrow$ enters TCA cycle.

• **Clinical Correlate:** Vitamin B12 deficiency or Mutase enzyme deficiency causes accumulation of methylmalonic acid, leading to severe **Methylmalonic Aciduria**.

2. Peroxisomal β -Oxidation (VLCFA Oxidation):

Fatty acids with carbon chains $\geq \text{C}^{22}$ cannot enter the carnitine shuttle and must be oxidized in peroxisomes. The first dehydrogenation step uses **Acyl-CoA Oxidase**, which transfers electrons directly to $\text{O}_2 \rightarrow$ **Hydrogen Peroxide (H_2O_2)** (does not generate ATP). H_2O_2 is broken down by **Catalase** to yield nascent oxygen [O] to kill bacteria or convert to H_2O and O_2 .

• **Zellweger Syndrome:** Congenital defect in peroxisomal biogenesis. Causes VLCFA accumulation in tissues, severe hypotonia, hepatomegaly, craniofacial dysmorphism, myelin destruction, and death in infancy. Adrenoleukodystrophy (ALD) is an X-linked defect in peroxisomal membrane VLCFA transporter ABCD1.

3. Unsaturated Fatty Acid Oxidation:

Naturally occurring double bonds are in the **cis** configuration, whereas β -oxidation enzymes only recognize the **trans** configuration. Oxidation requires two additional enzymes: **Isomerase** (converts cis- Δ^3 -Enoyl-CoA to trans- Δ^2 -Enoyl-CoA) and **Reductase** (removes double bonds in polyunsaturated fatty acids).

• **Energetics:** Bypasses the initial FADH_2 -generating Acyl-CoA dehydrogenase step, resulting in a **lower energy yield** compared to saturated fatty acids.

SPECIAL OXIDATIONS

α -Oxidation: Oxidation at α -carbon. For branched-chain fatty acids (e.g. phytanic acid). Requires Phytanoyl-CoA hydroxylase. Defect $\rightarrow$ **Refsum's disease**.

ω -Oxidation: Oxidation at terminal (ω) carbon. Minor pathway in ER. Produces dicarboxylic acids. Important in MCAD deficiency.

Peroxisomal β -Oxidation: For very long chain FA ($>\text{C}^{20}$). Generates H_2O_2 (catalase destroys it). No ATP produced directly. Defect $\rightarrow$ **Zellweger syndrome, Adrenoleukodystrophy (ALD)**.

03

Ketone Bodies

★ EXAMINER FAV

// KETOGENESIS – SYNTHESIS (IN LIVER MITOCHONDRIA)

2 × Acetyl-CoA

→ **Thiolase**

Acetoacetyl-CoA

↓ + Acetyl-CoA → **HMG-CoA Synthase**

RLS

HMG-CoA

↓ **HMG-CoA Lyase**

Acetoacetate

✓ ↓

β-Hydroxybutyrate
(NADH reduction)Acetone
(spontaneous decarboxylation)
→ *fruity breath*

Ketone bodies are made in LIVER but used in EXTRAHEPATIC tissues (brain, heart, muscle, kidney).
Liver lacks Succinyl-CoA transferase (thiophorase) – so it cannot use its own ketones.



Ketolysis (utilisation) – in extrahepatic tissues: β-Hydroxybutyrate → Acetoacetate (by HBD) →
Acetoacetyl-CoA (by Succinyl-CoA transferase, needs Succinyl-CoA) → 2 Acetyl-CoA → TCA → ATP.

Ketosis — Clinical Scenarios

- **Fasting Ketosis (Physiological / Safe):**

An overnight fast triggers a decrease in glucose → insulin falls, glucagon rises → activates Hormone Sensitive Lipase (HSL) to trigger mild adipose lipolysis → Free Fatty Acids (FFA) are released and transported to the liver → Liver β -oxidation increases, generating Acetyl-CoA. At the same time, Gluconeogenesis is activated, depleting Oxaloacetate (OAA) → Acetyl-CoA is diverted from TCA to Ketogenesis.

- **Safety Mechanism:** A **small amount of circulating insulin remains**, which acts as a molecular brake on lipolysis, preventing it from spiraling out of control. Ketones remain low (< 1 mg/dL) and blood pH remains stable and fully compensated.

- **Diabetic Ketoacidosis (DKA - Pathological / Unsafe):**

Seen in uncontrolled Type 1 Diabetes Mellitus. Insulin is **completely absent (zero insulin)** → lipolysis is thrown into massive, unchecked overdrive → FFA flooded into the liver → Acetyl-CoA production explodes. Because insulin is zero, glucose cannot enter cells, making the liver think the body is starving → Gluconeogenesis is extremely active, depleting OAA completely. Acetyl-CoA cannot enter TCA at all and is converted entirely to Ketones.

- **Clinical Presentation:** Acidic ketone bodies accumulate massively → Bicarbonate buffer system is overwhelmed → blood pH drops, causing severe **High Anion Gap Metabolic Acidosis**. Patients present with **Kussmaul breathing** (deep, rapid breathing to blow off CO_2), fruity acetone breath, profound dehydration, and coma.

- **Diagnostic & Treatment:** **Gerhardt's Test** is positive for primary acetoacetate. Treatment requires immediate administration of (1) **Glucose + Insulin** (insulin reverses lipolysis) and (2) **NaHCO_3** with aggressive electrolyte/fluid resuscitation.

- **DKA (Diabetic Ketoacidosis):** Severe — No insulin → uncontrolled lipolysis → massive FFA → overwhelming ketogenesis. Acetone breath, pH < 7.3 , High Anion Gap Metabolic Acidosis. Emergency.

- **Alcoholic ketoacidosis:** Ethanol → NADH $\uparrow$ → OAA → Malate (depletes OAA) → same result as starvation.

- **Key exam trap:** Both starvation AND DKA cause ketosis but **DKA is more severe** because insulin deficiency removes the brake on lipolysis completely.

04 Fatty Acid Synthesis (Lipogenesis)

// DE NOVO FATTY ACID SYNTHESIS – KEY STEPS

Acetyl-CoA (mitochondria)

→ **Citrate shuttle to cytoplasm**

Acetyl-CoA (cytoplasm)

↓ **Acetyl-CoA Carboxylase (ACC)** RLS + Biotin + CO₂ + ATP

Malonyl-CoA (3C)

↓ **Fatty Acid Synthase Complex (FAS)** – 7 rounds × 2C addition

Palmitate (C16:0) – first product

Location:

Cytoplasm
(+ SER for elongation)

Reducing agent:

NADPH (from HMP shunt
+ malic enzyme)

Cost per Palmitate:

7 ATP + 14 NADPH
+ 8 Acetyl-CoA

CYTOPLASMIC SHUTTLE & ACC REGULATION

De novo fatty acid synthesis occurs in the **cytosol** of the **liver and lactating mammary glands**. Mitochondrial Acetyl-CoA (produced from pyruvate or β-oxidation) cannot cross the inner mitochondrial membrane.

Citrate Shuttle:

Acetyl-CoA is condensed with Oxaloacetate to form **Citrate** (via Citrate Synthase), which exits to the cytoplasm via a tricarboxylate carrier. In the cytosol, **ATP Citrate Lyase** cleaves Citrate back into Acetyl-CoA and Oxaloacetate (consuming ATP).

ACC: The Rate-Limiting Step:

Cytoplasmic Acetyl-CoA is converted to Malonyl-CoA by **Acetyl-CoA Carboxylase (ACC)**, the rate-limiting enzyme. This carboxylation reaction requires three essential components: **ATP, Biotin (Vitamin B7), and CO₂** (as bicarbonate).

- **Allosteric Regulation:** Activated by **Citrate** (triggers polymerization of inactive monomers into active polymeric filaments) and inhibited by **Palmitoyl-CoA** (triggers depolymerization).
- **Hormonal Regulation:** **Insulin** dephosphorylates and activates ACC (+), promoting fat storage. **Glucagon and Epinephrine** phosphorylate and inhibit ACC (-) via AMP-activated protein kinase (AMPK) and PKA, shutting down fat synthesis during starvation/stress.

FATTY ACID SYNTHASE (FAS) HOMODIMER COMPLEX

The remaining steps of lipogenesis are catalyzed by **Fatty Acid Synthase (FAS)**, a multi-enzyme, multi-functional polypeptide complex that functions as a head-to-tail **homodimer**.

Structural Components:

Each monomer contains **7 distinct catalytic domains** and an **Acyl Carrier Protein (ACP)** domain. ACP is loaded with a **Phosphopantetheine** (derived from Vitamin B5) prosthetic group.

Active Sulfhydryl (-SH) Groups:

FAS contains two essential -SH groups that must work in close circular coordination:

1. **Cysteine-SH** group on the β -Ketoacyl Synthase (KS) enzyme domain of one monomer.
2. **Phosphopantetheine-SH** group on the ACP domain of the adjacent monomer.

The 4-Step Cycle (CRDR Mnemonic):

Malonyl-CoA units are transferred to the ACP, and the growing acyl chain undergoes a repeating 4-step sequence:

1. **Condensation (C)**: Condensation of malonyl-ACP with the acetyl group on Cysteine-SH, releasing CO_2 (catalyzed by Ketoacyl Synthase).
2. **Reduction (R)**: Reduction of the β -keto group to a β -hydroxy group, using **NADPH** (catalyzed by Ketoacyl Reductase).
3. **Dehydration (D)**: Dehydration of the β -hydroxy acyl chain to form a double bond, releasing **H²O** (catalyzed by Dehydratase).
4. **Reduction (R)**: Reduction of the double bond to a saturated single bond, using **NADPH** (catalyzed by Enoyl Reductase).

This 4-step cycle repeats **7 times** (adding 2 carbons per cycle) until the 16-carbon **Palmitate** is synthesized, which is cleaved from the complex by **Thioesterase**.



ACC (Acetyl-CoA Carboxylase)

RLS

— Activated by: Citrate, Insulin. Inhibited by: Palmitoyl-CoA (product inhibition), Glucagon, Epinephrine (via phosphorylation by PKA). **Malonyl-CoA** — the product — also inhibits CPT-I, preventing simultaneous FA synthesis + breakdown.

Synthesis vs β -Oxidation

Location: **Cytoplasm** vs **Mitochondria**

Carrier: **ACP (Synthesis)** vs **CoA (Oxidation)**

NADPH consumed vs NADH+FADH₂ produced

Citrate shuttle IN vs Carnitine shuttle IN

Stereochemistry: **L** vs **D** intermediate

Elongation & Desaturation

Elongation: ER + mitochondria (Malonyl-CoA units)

Desaturation: ER $\rightarrow$ Δ^9 -desaturase (most common)

Humans: **cannot introduce double bonds beyond C9**

$\therefore$ Linoleic (ω_6) + α -Linolenic (ω_3) = **Essential FA**

EFA deficiency $\rightarrow$ scaly skin, $\uparrow$ susceptibility to infection

05

Cholesterol Synthesis

★ EXAMINER FAV

// CHOLESTEROL SYNTHESIS – KEY STEPS (EXAM CONDENSED)

3 Acetyl-CoA

→ **HMG-CoA Synthase (cytoplasm)**

HMG-CoA

↓ **HMG-CoA Reductase** RLS + 2 NADPH**Mevalonate (6C)** ← Statin target↓ → **Isopentenyl pyrophosphate (IPP, 5C)** → **Squalene (30C)**

Lanosterol

→ **19 steps****Cholesterol (27C)****Location:**Cytoplasm + ER
(mainly liver)**ACAT:**Esterifies cholesterol
for storage**LCAT:**Esterifies cholesterol
in plasma (HDL)

DETAILED MOLECULAR MECHANISM: HOW STATINS LOWER BLOOD LDL

Statins (e.g., Atorvastatin, Rosuvastatin) are structural analogues of HMG-CoA and act as potent **competitive inhibitors** of the rate-limiting enzyme **HMG-CoA Reductase**.

Step-by-Step Liver-to-Blood Response:

- 1 Active Site Competition:** Statins bind directly to the active site of HMG-CoA Reductase, preventing the conversion of HMG-CoA to Mevalonate.
- 2 Intracellular Cholesterol Drop:** Synthesis of cholesterol plummets inside hepatocytes.
- 3 SREBP-2 Pathway Activation:** Low intracellular cholesterol causes the transcription factor **SREBP-2** (Sterol Regulatory Element Binding Protein-2) to migrate to the nucleus.
- 4 LDL Receptor Upregulation:** SREBP-2 upregulates the transcription and synthesis of **LDL Receptors** on the hepatocyte cell membrane.
- 5 LDL Blood Clearance:** The increased density of LDL receptors pulls LDL particles (Apo B-100 containing) out of the bloodstream.
- 6 Serum Drop:** Circulating blood LDL level drops dramatically, lowering cardiovascular plaque risk.

**HMG-CoA Reductase**

RLS

— Inhibited by: Statins (competitive), High intracellular cholesterol (via SREBP-2 pathway), Glucagon. Activated by: Insulin. **Statins** (Atorvastatin, Rosuvastatin) → ↓ cholesterol → upregulate LDL receptors → ↑ LDL clearance from blood.

PRODUCTS OF CHOLESTEROL

1Bile acids (primary: cholic, chenodeoxycholic) — essential for fat digestion

2Steroid hormones — glucocorticoids, mineralocorticoids, sex hormones

3Vitamin D — 7-dehydrocholesterol → D3 (skin, UV light)

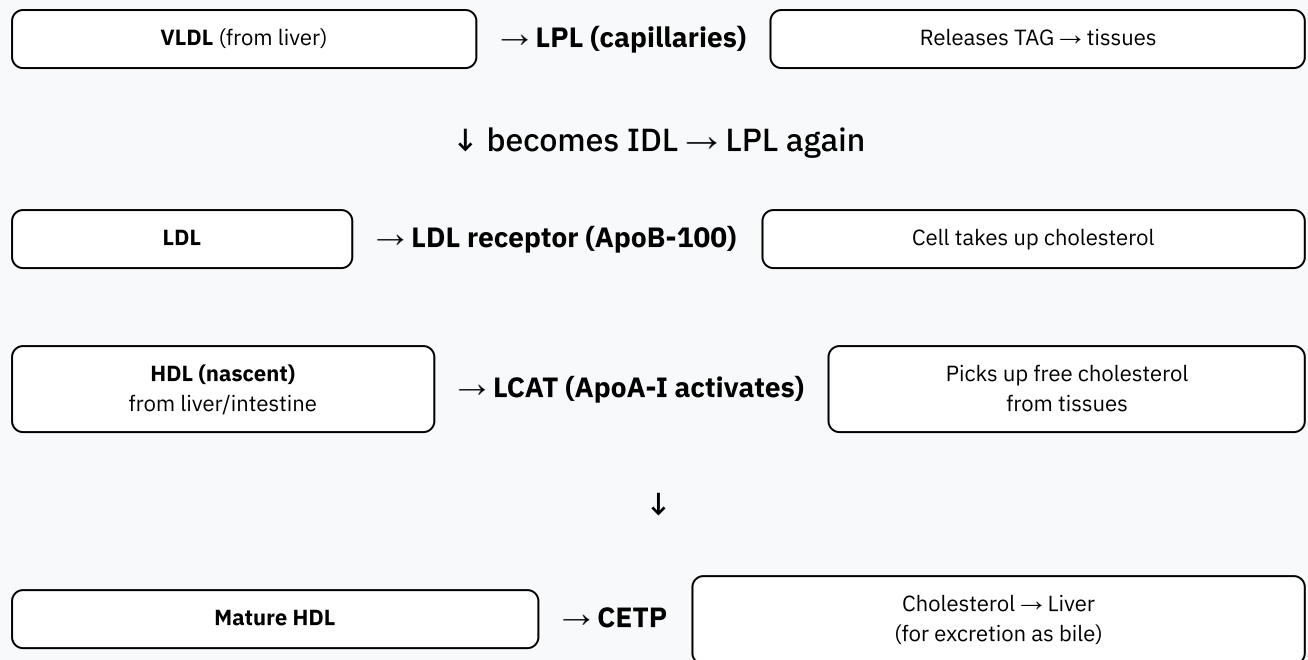
4Cell membrane — modulates fluidity

06 **Plasma Lipoproteins**

★ EXAMINER FAV

LIPOPROTEIN	ORIGIN	MAIN LIPID	KEY APO	FUNCTION
CHYLOMICRON	Intestine	TAG (85%)	ApoB-48, ApoC-II, ApoE	Transport dietary TAG to tissues
VLDL	Liver	TAG (55%)	ApoB-100, ApoC-II, ApoE	Transport endogenous TAG
IDL	VLDL remnant	TAG + Chol	ApoB-100, ApoE	Intermediate; → LDL
LDL	IDL (VLDL remnant)	Cholesterol (45%)	ApoB-100	Delivers cholesterol to cells — "Bad"
HDL	Liver + Intestine	Protein (50%)	ApoA-I, ApoA-II	Reverse cholesterol transport — "Good"

// VLDL METABOLISM & REVERSE CHOLESTEROL TRANSPORT



LPL vs. HSL (Classic Viva Question): Lipoprotein Lipase (LPL) sits on capillaries, is activated by Insulin and ApoC-II, and clears TAGs from the blood into tissues (Fed state). Hormone Sensitive Lipase (HSL) sits inside adipocytes, is activated by Glucagon/Epinephrine, and breaks down stored fat to release free fatty acids into the blood (Fasting state).

♥ TANGIER DISEASE (ABCA1 TRANSPORTER DEFECT)

Tangier Disease is an autosomal recessive disorder caused by mutations in the **ABCA1 transporter** protein.

Pathophysiology:

ABCA1 is responsible for pumping free cholesterol out of cells onto lipid-poor Apo A-I (nascent discoidal HDL). Without functional ABCA1, cells cannot export cholesterol → HDL cannot mature and is rapidly degraded in kidneys.

Clinical Presentation:

- **Zero HDL** in blood (or extremely low levels).
- Cholesterol esters accumulate in the macrophages of the reticuloendothelial system.
- **Massively enlarged bright orange tonsils** (hallmark sign, packed with cholesterol-laden macrophages).
- Severe hepatosplenomegaly, peripheral neuropathy, and highly increased risk of premature **Atherosclerosis**.

❤ LIPOPROTEIN LIPASE (LPL) KM RULE & TISSUE AFFINITIES

Lipoprotein Lipase (LPL) is anchored to capillaries to hydrolyze TAGs in Chylomicrons and VLDLs. Different tissues express different LPL isoforms with distinct kinetic properties:

Cardiac Muscle LPL: Low Km (High Affinity)

The heart relies constantly on fatty acid oxidation for fuel. The low Km of cardiac LPL ensures that the heart can extract TAGs for energy even when circulating blood lipid levels are extremely low.

Adipose Tissue LPL: High Km (Low Affinity)

Adipose tissue LPL is regulated by insulin and active only in the postprandial (fed) state. Its high Km ensures that fat storage occurs only when there is an abundance of blood lipids.



Familial Hypercholesterolaemia (FH): LDL receptor defect (ApoB-100 mutation) → LDL not cleared → ↑ serum LDL → premature atherosclerosis + xanthomas. Heterozygous: LDL ~2× normal. Homozygous: LDL ~6× normal, MI before age 20.

07 Phospholipids & Glycolipids

Key Phospholipids

Lecithin (PC) — most abundant; cell membrane + lung surfactant

Cephalin (PE) — cell membranes; coagulation

Sphingomyelin — myelin sheath; SM:PC ratio used for fetal lung maturity

Cardiolipin — inner mitochondrial membrane; antigen in syphilis test (VDRL)

Plasmalogen — heart + brain; ether phospholipid

Phospholipases

PLA₂ — releases arachidonic acid → eicosanoids (PGs, TXA₂, LTs)

PLC — generates DAG + IP₃ (2nd messengers)

PLD — generates phosphatidic acid

Cobra venom → PLA₂ → haemolysis via lysolecithin



Surfactant — Classic PYQ (2015-S, 2019-S)

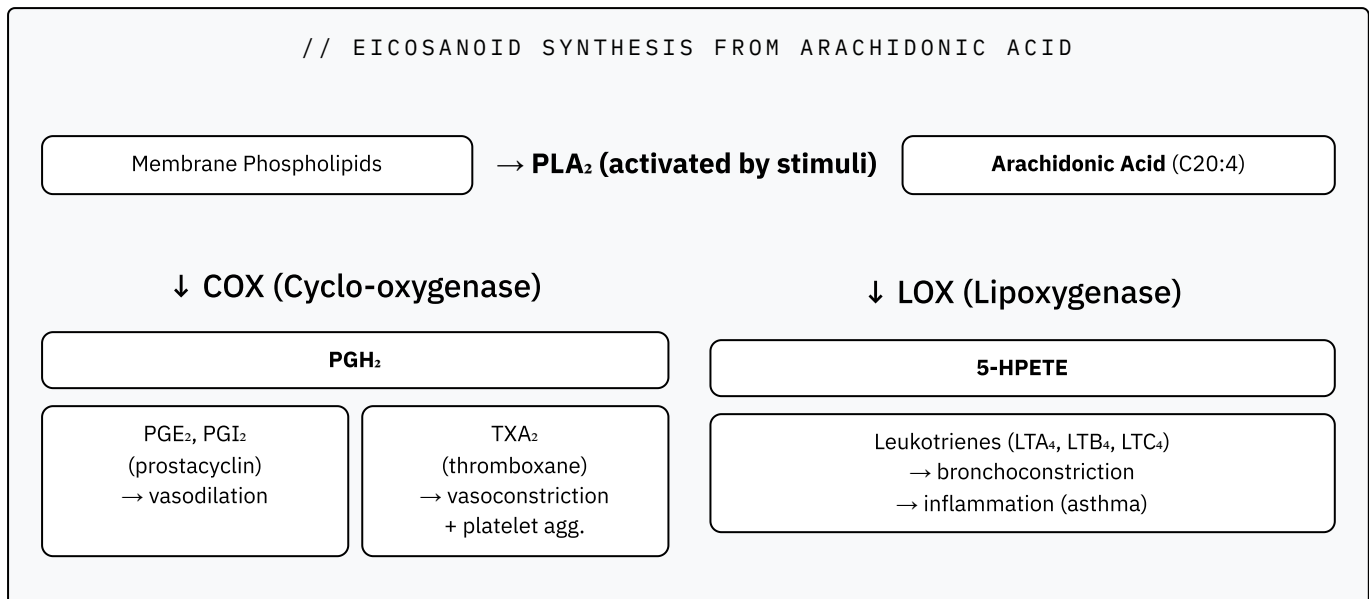
- **Dipalmitoyl lecithin (DPPC)** = lung surfactant. Produced by Type II pneumocytes. Reduces surface tension in alveoli at end-expiration → prevents alveolar collapse.
- **Respiratory Distress Syndrome (RDS) / Hyaline Membrane Disease:** Premature infants (<35 weeks) — surfactant deficiency → alveolar collapse → respiratory failure. Maternal corticosteroids accelerate surfactant synthesis.
- **L:S ratio** (Lecithin:Sphingomyelin) in amniotic fluid — >2 = mature fetal lungs.

Lipid Storage Diseases (Sphingolipidoses) — High Yield

- **Tay-Sachs Disease:** Hexosaminidase A deficiency → Ganglioside GM2 accumulates → Cherry-red spot on macula, neurodegeneration, no hepatosplenomegaly.
- **Gaucher's Disease:** Glucocerebrosidase deficiency → Glucocerebroside accumulates → Crumpled tissue paper macrophages, hepatosplenomegaly, bone crises. (Most common).
- **Niemann-Pick Disease:** Sphingomyelinase deficiency → Sphingomyelin accumulates → Cherry-red spot on macula + hepatosplenomegaly (foam cells).

08 Eicosanoids

// EICOSANOID SYNTHESIS FROM ARACHIDONIC ACID



COX inhibitors: Aspirin — irreversible acetylation of COX (serine residue). NSAIDs — reversible competitive inhibition. COX-2 selective inhibitors (Celecoxib) → reduced GI side effects. **Aspirin's anti-platelet effect persists 7–10 days** (platelet lifespan) because platelets have no nucleus — cannot synthesise new COX.

09 Fatty Liver & Lipotropic Factors



DEFINITION

Fatty liver (hepatic steatosis) = accumulation of TAG in hepatocytes >5% of liver weight. Results from imbalance between TAG synthesis and VLDL export.

// CAUSES OF FATTY LIVER

↑ FA delivery to liver
(starvation, obesity, DM)

↑ FA synthesis
(high carb diet, insulin)

↓ FA oxidation
(alcohol, hypoxia)

↓ Apoprotein synthesis
(protein malnutrition, CCl₄)

↓ Lipotropic factors
(↓ choline, methionine)

↓ All lead to:

TAG accumulates in hepatocytes → Fatty Liver



Lipotropic factors prevent fatty liver by promoting VLDL synthesis/export: **Choline** (for phosphatidylcholine in VLDL), **Methionine** (for choline synthesis via transmethylation), **Inositol**, Betaine, Vitamin B12, Folate. Deficiency → fat cannot be exported → accumulates.

10 Memory Mnemonics

● MNEMONIC 1 – B-OXIDATION 4 STEPS

"Elephants Have Thick Hides" → EHTH

Enoyl-CoA dehydrogenase (FAD) → Hydratase (H₂O) → Thiohydroxyacyl-CoA dehydrogenase (NAD) → Thiolase (CoA) — products: FADH₂, NADH, Acetyl-CoA each round.

● MNEMONIC 2 – LIPOPROTEIN DENSITY ORDER

"Chubby Vikings Inspire Lovely Humans"

Chylomicron → VLDL → IDL → LDL → HDL

Density increases (↑ protein, ↓ fat): Chylomicron = least dense, HDL = most dense.

● MNEMONIC 3 – KETONE BODY NAMES

"A Beta-Blocker Acetates" → ABA

Acetoacetate · Beta-hydroxybutyrate · Acetone — three ketone bodies. Only first two carry energy; acetone is exhaled (fruity breath in DKA).

● MNEMONIC 4 – APOLIPOPROTEINS AND FUNCTION

"A-I Activates LCAT; B-100 Binds LDL-R; C-II Calls LPL; E Everyone enters"

ApoA-I → activates LCAT (HDL maturation) · ApoB-100 → binds LDL receptor · ApoC-II → activates Lipoprotein Lipase (LPL) · ApoE → receptor-mediated uptake of remnants.

● MNEMONIC 5 – LIPOTROPIC FACTORS

"Clever Mice In Big Labs"

Choline · Methionine · Inositol · B12 · Lecithin — deficiency of any → fatty liver.

11 Clinical Correlations

🔗 High-Yield Clinical Links

- **Statin therapy:** Inhibit HMG-CoA reductase → ↓ cholesterol synthesis → upregulate hepatic LDL receptors → ↑ LDL clearance. Side effect: myopathy (↑ CK-MM). Monitor liver enzymes.
- **MCAD deficiency:** Medium Chain Acyl-CoA Dehydrogenase deficiency → β-oxidation block → hypoketotic hypoglycaemia in fasting infants → **SIDS-like presentation**. Increased C8-C10 acylcarnitines on newborn screen.
- **Alcohol → Fatty liver:** Ethanol → Acetaldehyde → NADH ↑ ↑ → NADH/NAD⁺ ratio ↑ → OAA → Malate (depletes TCA) → FA synthesis ↑ + FA oxidation ↓ → TAG accumulates. Also: Acetaldehyde inhibits VLDL secretion.
- **Refsum's disease:** Phytanoyl-CoA hydroxylase deficiency → phytanic acid accumulates → peripheral neuropathy, retinitis pigmentosa, cerebellar ataxia. Avoid chlorophyll-rich foods.
- **Tangier disease:** ApoA-I deficiency → HDL deficiency → cholesterol accumulates in macrophages (foam cells) → orange tonsils, hepatosplenomegaly, ↑ atherosclerosis risk.
- **Abetalipoproteinaemia:** ApoB deficiency → cannot form chylomicrons + VLDL → fat malabsorption, acanthocytes, ataxia, retinitis pigmentosa. No dietary fat or fat-soluble vitamins absorbed.

VIVA RAPID FIRE – LIPID METABOLISM

Q1 Why can't even-chain fatty acids contribute to net gluconeogenesis?

Even-chain FA → β -oxidation → Acetyl-CoA only. Acetyl-CoA cannot be converted back to pyruvate or OAA in net amounts because the Pyruvate Dehydrogenase reaction is irreversible. Acetyl-CoA entering TCA condenses with OAA → both carbons are released as CO_2 in the cycle – no net carbon gain for GNG. Exception: odd-chain FA → Propionyl-CoA → Succinyl-CoA → can enter GNG.

Q2 Why does the liver produce but not use ketone bodies?

Liver lacks Succinyl-CoA transferase (3-oxoacid CoA transferase / thiophorase) – the enzyme needed for the first step of ketolysis. This is actually physiologically appropriate: the liver produces ketone bodies specifically to export fuel to the brain and other tissues during starvation. If the liver used its own ketones, this export function would fail.

Q3 How does Malonyl-CoA coordinate FA synthesis and oxidation?

Malonyl-CoA is the first committed product of FA synthesis (made by ACC). It simultaneously inhibits CPT-I – the gatekeeper for FA entry into mitochondria for β -oxidation. So when FA synthesis is ON (high insulin, fed state), malonyl-CoA rises and blocks β -oxidation. When fasting → glucagon → ↓ malonyl-CoA → β -oxidation proceeds. One molecule coordinates two opposing pathways.

Q4 Calculate ATP yield from complete oxidation of palmitate (C16:0).

Activation: -2 ATP (Acyl-CoA Synthetase). 7 rounds of β -oxidation: 7 FADH_2 ($7 \times 1.5 = 10.5$ ATP) + 7 NADH ($7 \times 2.5 = 17.5$ ATP) = 28 ATP. 8 Acetyl-CoA → TCA: $8 \times 10 = 80$ ATP. Total = $80 + 28 - 2 = 106$ ATP (modern P/O ratios). Using older textbook values (3 per NADH, 2 per FADH_2): $8 \times 12 + 7 \times 5 - 2 = 129$ ATP. Match whichever your professor uses.

Q5 What is reverse cholesterol transport and why is HDL "good"?

Reverse cholesterol transport is the process by which HDL picks up excess free cholesterol from peripheral tissues and transports it back to the liver for excretion as bile acids. Step 1: Nascent HDL (disc-shaped) secreted by liver/intestine. Step 2: ApoA-I activates LCAT → esterifies cholesterol → HDL matures (spherical). Step 3: Mature HDL delivers cholesterol ester to liver via SR-B1 receptor OR transfers via CETP to LDL/VLDL. This removes cholesterol from artery walls → anti-atherogenic = "good."

Q6 Why does alcohol cause fatty liver?

Ethanol → acetaldehyde (ADH) → acetate (ALDH). Both steps produce NADH. High NADH/ NAD^+ ratio: (1) Inhibits β -oxidation (NAD^+ depleted). (2) OAA → Malate (TCA depleted → ↓ FA oxidation). (3) Acetaldehyde directly inhibits VLDL secretion. (4) ↑ Malonyl-CoA → ↑ FA synthesis. Net: FA cannot leave liver → TAG accumulates → fatty liver → if chronic, cirrhosis.

Q7 Distinguish between mitochondrial and peroxisomal β -oxidation.

Mitochondrial: substrate = C4–C20 FA; entry via carnitine shuttle; produces ATP directly ($\text{NADH} + \text{FADH}_2$ feed ETC); fully oxidises to Acetyl-CoA. Peroxisomal: substrate = VLCFA (>C20) and branched-chain FA; no carnitine shuttle needed; first oxidation uses FAD → H_2O_2 (neutralised by catalase); does NOT produce ATP directly – electrons go to O_2 not ETC; incompletely oxidises to medium-chain acyl-CoA for mitochondrial completion. Defect in peroxisomal → Zellweger, ALD.

Q8 Why is dipalmitoyl lecithin called the surfactant and what is its clinical significance?

Dipalmitoyl phosphatidylcholine (DPPC) has two saturated palmitate tails – this makes it pack tightly at the air-water interface of alveoli, dramatically reducing surface tension. By LaPlace's law, lower surface tension means smaller pressure required to keep alveoli open. Produced by Type II pneumocytes – develops fully only by ~35 weeks gestation. Premature birth before this → RDS (Hyaline Membrane Disease). Antenatal corticosteroids accelerate maturation. L:S ratio >2 in amniotic fluid = lung maturity.

13 Group-D EQ Template

📄 HOW TO ANSWER: "HDL IS GOOD CHOLESTEROL" / "STATIN DRUGS ACT AS CHOLESTEROL-LOWERING AGENTS"

Use the flowchart formula — Define → Mechanism → Kinetics/Regulation → Clinical consequence.

// EQ ANSWER FLOWCHART — "STATINS ACT AS CHOLESTEROL-LOWERING AGENTS"

Define
Competitive inhibitors of HMG-CoA reductase



Mechanism
Statin → ↓ Mevalonate → ↓ Cholesterol synthesis in hepatocytes



Compensatory upregulation
↓ Intracellular cholesterol → SREBP-2 activated → ↑ LDL receptor expression



Clinical outcome
↑ LDL receptor → ↑ clearance of LDL from plasma → ↓ serum LDL



Side effect
Myopathy (↑ CK-MM) — statin-induced myositis

MBBS Biochemistry — Module 03: Lipid Metabolism

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